

gemstone-rs Codegen Profile Schema

Project profile JSON validation for explorer and VS Code workflows



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Codegen Profile Schema

`gemstone-rs` project profile files make explorer and VS Code codegen workflows repeatable. The default file name is:

```
gemstone-rs.codegen-profiles.json
```

This repository includes a sample:

```
examples/codegen/gemstone-rs.codegen-profiles.json
```

The JSON Schema is checked in at:

```
schemas/gemstone-rs.codegen-profiles.schema.json
```

Validate the sample from a checkout:

```
cargo run -p gemstone-rs-cli -- profile sample
cargo run -p gemstone-rs-cli -- profile init gemstone-rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile validate examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile validate --json examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile list examples/codegen/gemstone-rs.codegen-
profiles.json
cargo run -p gemstone-rs-cli -- profile show default examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile resolve default examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- profile check examples/codegen/gemstone-rs.codegen-
profiles.json
cargo run -p gemstone-rs-cli -- profile check --json examples/codegen/gemstone-
rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- codegen check-profile default examples/codegen/
gemstone-rs.codegen-profiles.json
cargo run -p gemstone-rs-cli -- codegen explain-profile --json default examples/
codegen/gemstone-rs.codegen-profiles.json
```

For an installed CLI:

```
gemstone-rs profile sample
gemstone-rs profile init gemstone-rs.codegen-profiles.json
gemstone-rs profile validate gemstone-rs.codegen-profiles.json
gemstone-rs profile validate --json gemstone-rs.codegen-profiles.json
```

```
gemstone-rs profile list gemstone-rs.codegen-profiles.json
gemstone-rs profile show default gemstone-rs.codegen-profiles.json
gemstone-rs profile resolve default gemstone-rs.codegen-profiles.json
gemstone-rs profile check gemstone-rs.codegen-profiles.json
gemstone-rs profile check --json gemstone-rs.codegen-profiles.json
gemstone-rs codegen check-profile default gemstone-rs.codegen-profiles.json
gemstone-rs codegen explain-profile --json default gemstone-rs.codegen-profiles.json
```

Expected output:

```
profile ok: examples/codegen/gemstone-rs.codegen-profiles.json (3 profiles: default,
object-wrapper, bridge-mapping)
profile check: examples/codegen/gemstone-rs.codegen-profiles.json (3 profiles)
summary: 3 ok, 0 stale, 0 errors, 3 total
ok      default config=examples/codegen/gemstone-rs.codegen      output=examples/
codegen/generated/gemstone_wrappers.rs
```

The JSON form includes aggregate fields so CI and editor integrations can fail with a concise summary:

```
{"ok":true,"profileCount":3,"okCount":3,"staleCount":0,"errorCount":0}
```

Shape

```
{
  "kind": "gemstone-rs-explorer-codegen-profiles",
  "version": 1,
  "profiles": [
    {
      "name": "default",
      "config": "examples/codegen/gemstone-rs.codegen",
      "root": "",
      "mapped": "BookingDraft",
      "className": "Object"
    }
  ]
}
```

Top-level fields:

- `kind` must be `gemstone-rs-explorer-codegen-profiles`.
- `version` must be `1`.
- `profiles` must be an array.

Profile fields:

- `name` is required, string-valued, non-empty, and unique.

- `config` is optional and string-valued.
- `root` is optional and string-valued.
- `mapped` is optional and string-valued.
- `className` is optional and string-valued.

Unknown top-level or profile fields are rejected. Non-string profile fields are rejected.

VS Code

Use `GemStone RS: Show Sample Project Profiles` to open the built-in sample in an untitled JSON editor, `GemStone RS: Create Project Profiles` to write the sample to a project file, and `GemStone RS: Validate Project Profiles` to run the same CLI validator and show the result in the GemStone RS output panel. `GemStone RS: List Project Profiles`, `GemStone RS: Show Project Profile`, and `GemStone RS: Resolve Project Profile` render parsed profile details through the same CLI parser, so you can check the active `config`/`root`/`mapped`/`class` fields and resolved config path without opening the explorer. Profile-specific commands use a QuickPick populated from the profile file. The workbench also contributes JSON validation for files named `gemstone-rs.codegen-profiles.json`, using the packaged schema copy in `vscode-gemstone-rs-workbench/schemas/`.

Without the extension, associate the schema with the profile file in VS Code:

```
{
  "json.schemas": [
    {
      "fileMatch": ["gemstone-rs.codegen-profiles.json"],
      "url": "./schemas/gemstone-rs.codegen-profiles.schema.json"
    }
  ]
}
```

Explorer Safety

The explorer validates project profile JSON before saving. It also rejects write paths containing `..` after URL decoding. Relative profile writes stay inside the configured `--codegen-root`; absolute write targets require `--allow-absolute-write-paths`.

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